

Karthikheyaa Kurra

Research Software Engineer

📍 Newark, DE ✉ karth@udel.edu 🔗 karthikheyaa 🌐 khey17

Summary

Research Software Engineer in the Department of Mathematical Sciences at the University of Delaware, working on scientific machine learning for partial differential equations (PDEs). My current research focuses on building hybrid solvers that combine neural networks with classical numerical methods.

Experience

University of Delaware, Research Software Engineer

Department of Mathematical Sciences, Newark, DE

- Developing a learned Hierarchical Poincaré–Steklov solver for semilinear elliptic PDEs with significant speedups over traditional solvers while maintaining controlled accuracy. Jan 2024 – present
- Building hybrid solvers that combine neural networks with classical numerical methods for scientific machine learning. 2 years 7 months

research interests

Scientific Machine Learning: Operator Learning, Physics-Informed Neural Networks, Equivariant Neural Networks

Additional Interests: Computer Vision, Computational Optimal Transport